



# ZENITH AERO: GRAND VOYAGER

## PROJECT DOSSIER & INVESTMENT PROPOSAL

305cm Hybrid Artisan Pedicab — Long-Wheelbase Edition

### 1. DESIGN VISION: LUXURY SUSTAINABILITY

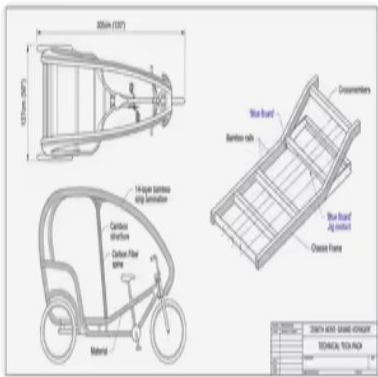
The **Zenith Aero: Grand Voyager** is a premium, carbon-reinforced bamboo vehicle designed for the luxury resort and eco-tourism market. By combining organic vibration-damping with aerospace-grade composite reinforcement, we provide a "floating ride" that steel frames cannot replicate.

#### LIFESTYLE CONCEPT: THE EXPERIENCE

#### PRIMARY ARTISTIC RENDER

### 2. CHASSIS ENGINEERING & DIMENSIONS

The chassis features an elongated **305cm (120")** wheelbase. This design ensures maximum passenger-to-driver separation, providing a truly high-end experience for two adult passengers.

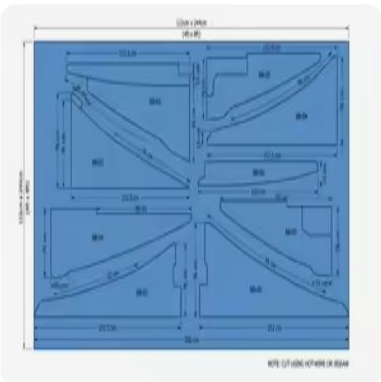


#### CORE SPECIFICATIONS

- **TOTAL LENGTH:** 305cm (120")
- **ARCHITECTURE:** 14-Layer Artisan Bamboo / Carbon Fiber Hybrid
- **LOAD CAPACITY:** 250kg (550 lbs)
- **WEIGHT:** Est. 75kg (Electrified Hybrid Construction)

### 3. ARTISAN FABRICATION: "BLUE BOARD" MOLDING

We utilize a low-CAPEX, high-precision molding method using high-density XPS foam templates. This allows for complex, aerodynamic curves without the expense of large-scale CNC tooling.



FABRICATION METHODOLOGY

- 1. **STRIP-BY-STRIP LAYUP:** Utilizing unbroken bamboo strips from Coolite poles.
- 2. **INTERNAL CARBON SPINE:** 3K UD Carbon Fiber tape embedded for longitudinal rigidity.
- 3. **VACUUM-SHRINK COMPRESSION:** 360-degree pressure via industrial heat-shrink tubing.

4. SCULPTED JOINTS & FINISH QUALITY

Critical joints are "overbuilt" with carbon fiber and structural epoxy, then sculpted by hand into an organic, aerodynamic silhouette.



FINISHING PROTOCOL

- **JOINT SCULPTING:** Utilizing Dremel rotary tools with carbide burrs for aero-transitions. [Mthr](#)
- **SANDING:** 80-grit "Shoe-Shine" method for perfectly rounded profiles. [Bona](#)
- **SEALING:** Multiple layers of Super Sap CCR Slow bio-epoxy for a glass-like waterproof finish. [Swellcomposites](#)

5. VANCOUVER SUPPLY CHAIN PARTNERS

The project leverages a verified, local BC supply chain, minimizing logistics risks and lead times.

| COMPONENT        | LOCAL PARTNER                | KEY ADVANTAGE                                                         |
|------------------|------------------------------|-----------------------------------------------------------------------|
| BAMBOO & RATTAN  | Coolite Bamboo               | Premium treated poles & artisan weaving <a href="#">Coolitebamboo</a> |
| COMPOSITES & XPS | Industrial Plastics & Paints | Carbon fiber & Blue Foam supplies <a href="#">Goindustrial</a>        |
| BIO-RESRINS      | Swell Composites             | Canadian source for Entropy Resins <a href="#">Swellcomposites</a>    |

INVESTMENT OPPORTUNITY

- **40% WEIGHT REDUCTION** compared to traditional steel pedicabs.
- **PASSIVE SUSPENSION:** Suspended rattan seating for industry-leading comfort.
- **MARKET READY:** Design validated by UBC academic faculty.

 Want more tailored results? [Add preference](#)